



United International University (UIU)
Department of Computer Science and Engineering
CSE 1325: DIGITAL LOGIC DESIGN, Final Exam, Spring 2025
Total Marks: 40, Duration: 2 hours
Answer ALL Questions

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules

1. Imagine you're developing a smart traffic monitoring system for an autonomous city. The system monitors a stream of binary signals transmitted by vehicles as they pass a specific checkpoint. Each vehicle's identifier is encoded as a binary sequence. The system must trigger a high-priority alert signal (1) if and only if it detects the identifier sequence "0110" in the incoming bit stream. If the pattern isn't found, the system continues to send 0 to the alert output. Do the following. (Use **D flip-flops** for the tasks.)

- A. Design the state diagram for the sequence recognizer. [2]
- B. Create a state table for the circuit. [2]
- C. Derive the minimized logic equations for the flip-flops inputs and output. [3]
- D. Draw the circuit diagram for the sequence recognizer. [1]

2. It's your birthday, and you've booked the entire Yum Cha District restaurant to celebrate with your friends. The restaurant uses a robotic serving arm controlled by a **3-bit counter** with state variables Q_2, Q_1, Q_0 to automate food service. You've arranged for a special menu where each of your guests will be served dishes in a fixed order:

Wonton in Chili Oil (010) $\rightarrow$ *Seafood Soup* (101) $\rightarrow$ *Prawn Tempura* (100) $\rightarrow$ *YCD Special Fried Rice* (110) $\rightarrow$ *Iced Tea* (011).

Once Iced Tea is served, the robot resets and starts the sequence again from *Wonton in Chili Oil* for the next guest. Ensure any unused states (e.g., 000, 001, 111) are treated as don't-cares. Do the following:

- A. Prepare a state table using **T Flip-Flop** inputs (T_2, T_1, T_0). [3]
 - B. Minimize the functions of T Flip-Flop inputs using Karnaugh Map. [3]
 - C. Draw the circuit diagram using Flip-Flop and other basic gates. [2]
3. A. Gus Fring's underground lab needs a new security system. The vault will only open when the correct combination of four chemical sensors (A, B, C, D) detect the right purity levels. Walter White has to design the logic circuit, knowing that one wrong calculation could mean the difference between freedom and facing Gus's wrath. He can only use one **4:1 multiplexer** and other gates. How can Walter White design the following function F and implement it as a circuit? [4]

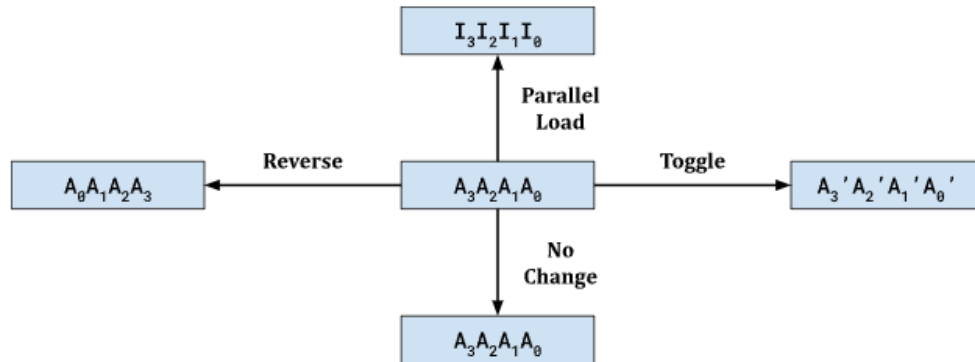
$$F(A, B, C, D) = \sum_m(1, 2, 4, 7, 8, 11, 13, 14)$$

- B. Implement the following functions using a single decoder. Note that the decoder should have an active low output. [4]

i) $F1(A, B, C) = \sum_m(1, 2, 5)$, ii) $F2(A, B, C) = \prod_M(0, 3, 7)$

4. You are designing a 4-bit register that controls a smart stick holding 4 LED bulbs. You provide two control switches—one for reverse (R) and one for toggle (T). When R is on exclusively (i.e., T is off), the register performs a bit-reversal. If T is on exclusively (i.e., R is off), all the bits are toggled. If both switches are off, the register holds its current value. If both are on, it loads four parallel inputs. The effect of each operation on the register bits is shown in the diagram below. Stored bits are denoted by A's, and the parallel inputs by I's.

[8]



Design the complete 4-bit register using D flip-flops, multiplexers, and any necessary logic gates.

5. You are part of a team managing a low-earth orbit satellite. The satellite receives various signals and messages, but it can only process one communication request at a time due to limited onboard computational and communication bandwidth. To ensure that the most critical signals are always handled first, you decide to implement a priority encoder. This encoder will take in multiple active signals and output a 3-bit binary code representing the highest-priority communication currently active.

Priority	Task	Input Symbol	Output Code
1 (Highest)	Critical System Alert	A	111
2	Command Signal	C	101
3	Instrument Data Transmission	D	001
4	Maintenance Log	M	011
5 (Lowest)	Background Sync	S	110

Complete the following steps:

- Derive the truth table of the priority encoder including the valid bit.
- Derive the Boolean expressions for all the outputs.
- Draw the logic diagram using basic gates.

[3]

[3]

[2]

Excitation Tables for Flip-Flops

Q(t)	Q(t+1)	T	Operation
0	0	0	No change
0	1	1	Complement
1	0	1	Complement
1	1	0	No change

Q(t)	Q(t+1)	D	Operation
0	0	0	Reset
0	1	1	Set
1	0	0	Reset
1	1	1	Set